REAR AXLE - 216RBI (Continued)

(8) Thread Pilot Stud C-3288-B into rear cover bolt hole below ring gear (Fig. 15).

(9) Attach a dial indicator C-3339 to Pilot Stud C-3288-B. Position the dial indicator plunger on a flat surface between the ring gear bolt heads (Fig. 15).

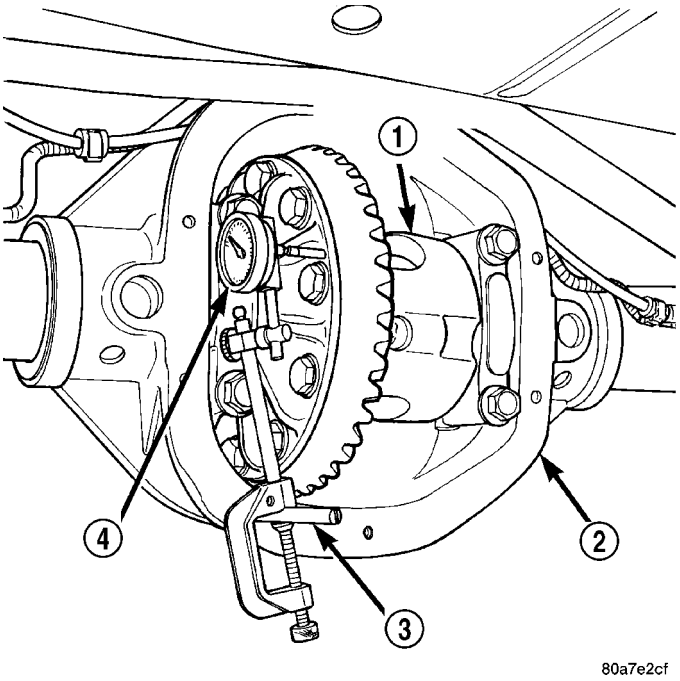

Fig. 15 DIFFERENTIAL SIDE PLAY MEASUREMENT

1 - DIFFERENTIAL CASE
2 - DIFFERENTIAL HOUSING
3 - PILOT STUD
4 - DIAL INDICATOR

(10) Push and hold differential case to pinion gear side of the housing and zero dial indicator (Fig. 16).

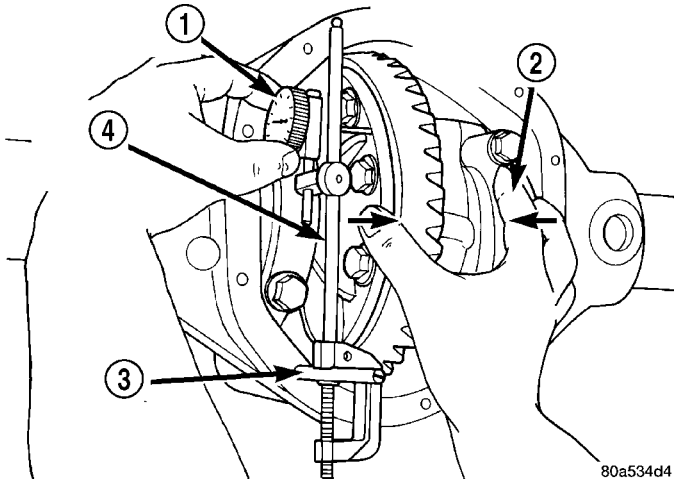

Fig. 16 ZERO DIAL INDICATOR

1 - DIAL INDICATOR
2 - FORCE DIFFERENTIAL CASE TO PINION GEAR SIDE
3 - PILOT STUD
4 - EXTENSION

(11) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 17).

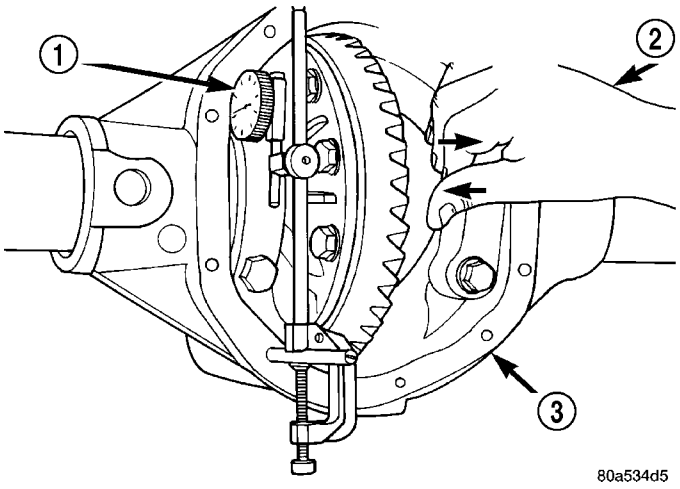

Fig. 17 RECORD DIAL INDICATOR READING

1 - READ DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - DIFFERENTIAL HOUSING

(12) Add dial indicator reading to the starting point shim thickness. This is the total shim thickness to achieve zero differential end play.

(13) Add 0.254 mm (0.010 in.) to the zero end play total. This new total represents the thickness of shims to compress, or preload the new bearings when the differential is installed.

(14) Rotate dial indicator out of the way on pilot stud.

(15) Remove differential case and dummy bearings from the housing.

(16) Install pinion gear in the housing. Install the pinion yoke and establish the correct pinion rotating torque.

(17) Install differential case and dummy bearings D-345 in the housing (without shims), install bearing caps and tighten bolts snug.

(18) Seat ring gear side dummy bearing (Fig. 14).

(19) Position the dial indicator plunger on a flat surface between the ring gear bolt heads. (Fig. 15).

(20) Push and hold differential case toward the pinion gear and zero dial indicator (Fig. 18).

(21) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 19).

(22) Subtract 0.05 mm (0.002 in.) from the dial indicator reading to compensate for backlash between ring and pinion gears. This total is the thickness shim required to achieve proper backlash.

(23) Subtract the backlash shim thickness from the total preload shim thickness. The remainder is the shim thickness required on the pinion side of the housing.